Zhuoyuan Li

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Institute for Functional Intelligent Materials
National University of Singapore

Education

2020–2025	**Ph.D.**, School of Mathematical Sciences, Peking University, Beijing, China co-supervisors: Prof. Pingwen Zhang[1], Prof. Bin Dong[2]
2016–2020	**B.Sc.**, School of Mathematics, Sichuan University, Chengdu, China

Working Experience

2025–present	Research Fellow	Institute for Functional Intelligent Materials National University of Singapore, Singapore

Research Interests

- Deep learning for multiscale modeling
- Numerical weather prediction and data assimilation
- Inverse problems and uncertainty quantification

Publications

Google Scholar

† → Equal contribution; $*$ → Corresponding author(s)

Journal Articles

J1. **Zhuoyuan Li**, Dong*, B. & Zhang*, P. State-observation augmented diffusion model for nonlinear assimilation with unknown dynamics. *Journal of Computational Physics* **539,** 114240. ISSN: 0021-9991. https://doi.org/10.1016/j.jcp.2025.114240 (2025).

J2. **Zhuoyuan Li***, Dong, B. & Zhang, P. Latent assimilation with implicit neural representations for unknown dynamics. *Journal of Computational Physics* **506,** 112953. ISSN: 0021-9991. https://doi.org/10.1016/j.jcp.2024.112953 (2024).

[1] homepage: https://www.math.pku.edu.cn/pzhang/en/
[2] homepage: http://faculty.bicmr.pku.edu.cn/~dongbin/

Peer-reviewed Conference Proceedings

C1. Yang[†], P., Feng[†], Y., Chen[†], Z., Wu, Y. & **Zhuoyuan Li**[*]. *Spend Wisely: Maximizing Post-Training Gains in Iterative Synthetic Data Bootstrapping* in *Advances in Neural Information Processing Systems* **38** (2025). https://arxiv.org/abs/2501.18962.

Preprints

P1. Li[†], L., **Zhuoyuan Li**[†], Li[†], S., Zhan, K., Gao[*], H., Chen[*], C. & Yang[*], L. In-context modeling as a retrain-free paradigm for foundation models in computational science. *arXiv preprint arXiv:2604.23098.* https://arxiv.org/abs/2604.23098 (2026).

P2. **Zhuoyuan Li**[†], Zhu[†], A. & Li[*], Q. Hypothesis-driven construction of mesoscopic dynamics. *arXiv preprint arXiv:2605.16211.* https://arxiv.org/abs/2605.16211 (2026).

P3. Huang[†], X., **Zhuoyuan Li**[†], Liu, H., Wang, Z., Zhou, H., Dong[*], B. & Hua, B. Learning to simulate partially known spatio-temporal dynamics with trainable difference operators. *arXiv preprint arXiv:2307.14395.* https://arxiv.org/abs/2307.14395 (2023).

Talks

CSIAM 2024	Oct. 24-27, 2024	"Latent assimilation with implicit neural representations for unknown dynamics" (Selected Poster)

Teaching

Peking University

Spring 2023	Assistant Instructor, Advanced Algebra (II)
Fall 2022	Assistant Instructor, Advanced Algebra (I)
Spring 2022	Assistant Instructor, Advanced Algebra (II)
Fall 2020	Teaching Assistant[3], Advanced Mathematics (C)

Please see my homepage for more details.

Other Experience

May 2024	China Meteorological Administration Tornado Key Laboratory (link, in Chinese) – location: Foshan, Guangdong, China – deploy a CNN-based model for tornado detection and classification
Summer 2018	research internship organized by MITACS – location: University of Alberta, Edmonton, AB, Canada – topic: multi-marginal optimal transport (advisor: Prof. Brendan Pass)

Last updated: May 18, 2026

[3] without teaching tasks